

README - Proiect PCLP3, Partea I

Popescu Petrut-Alin | Grupa 312 CA

1. Introducere

Proiectul implementeaza **Partea I**: generare sintetica de dataset, EDA si antrenare model de baza. Tipul problemei: **clasificare binara** — se prezice daca un pacient are risc cardiovascular ridicat (*Cardiovascular risk: Yes / No*).

Informatiile medicale folosite pentru a defini factorii de risc si pragurile din scorul de risc cardiovascular au fost preluate de pe site-ul Organizatiei Mondiale a Sanatatii (World Health Organization): <https://www.who.int/health-topics/cardiovascular-diseases>

2. Generarea sintetica a setului de date

2.1 Coloanele brute

Fisierul **data_frame_generation.py** construiesc 1000 de instante folosind numpy, cu distributii realiste. Sunt acoperite 3 tipuri de date: intregi (Age, Pulse, Cholesterol), reale (Corporal body fat, Sleeped hours in a week) si categoricale (Gender, Smoker, Family history, Cardiovascular risk).

```
age          = np.random.randint(16, 90)
sex          = np.random.choice(['Masculin', 'Feminin'], p=[0.5, 0.5])
body_fat     = np.round(np.random.normal(26, 5), 5) # distributie normala
pulse        = np.random.randint(60, 120)
cholesterol  = np.random.randint(100, 300)
smoker       = np.random.choice(['Yes', 'No'], p=[0.25, 0.75])
fam_hist     = np.random.choice(['Yes', 'No'], p=[0.2, 0.8])
sleep_h      = np.round(np.random.uniform(3,10) * 7, 1)
```

2.2 Variabila tinta prin scor de risc

Eticheta nu este aleatorie — deriva logic dintr-un scor compus bazat pe factori de risc reali. Daca scorul depaseste pragul 5, instanta e marcata cu risc ridicat:

```
base_risk = (
    (age > 60)          * 2.0 + (body_fat > 29)      * 1.5 +
    (pulse > 85)        * 1.25 + (cholesterol > 250) * 2.5 +
    (smoker == 'Yes')   * 3.0 + (fam_hist == 'Yes') * 2.0 +
    (sleep_h < 35)      * 1.0
)
risk = np.where(base_risk + noise > 5, 'Yes', 'No')
```

2.3 Anomalii si zgomot introduse deliberat

Pentru realism: **erori de masurare** la 10 instante pentru Pulse (200-270 bpm) si 10 pentru body fat (60-84%); **anomalii genetice** — 7 tineri sub 28 ani nefumatori primesc eticheta Yes, iar 7 varsnici peste 70 ani fumatori primesc No; **valori lipsa** ~5% la Cholesterol si Corporal body fat (NaN).

2.4 Split si salvare

Split stratificat cu sklearn: **700 instante train**, **300 instante test**. Exportate ca data_training.csv si data_testing.csv.

3. Analiza exploratorie a datelor (EDA)

EDA realizata in **data_eda.py** pentru ambele subseturi. Valorile lipsa sunt ~5% pe coloanele Cholesterol si Corporal body fat. Tratate prin imputare cu media (numeric) si modul (categoric), calculate pe train si aplicate pe test pentru a evita data leakage.

3.1 Distributia variabilelor numerice

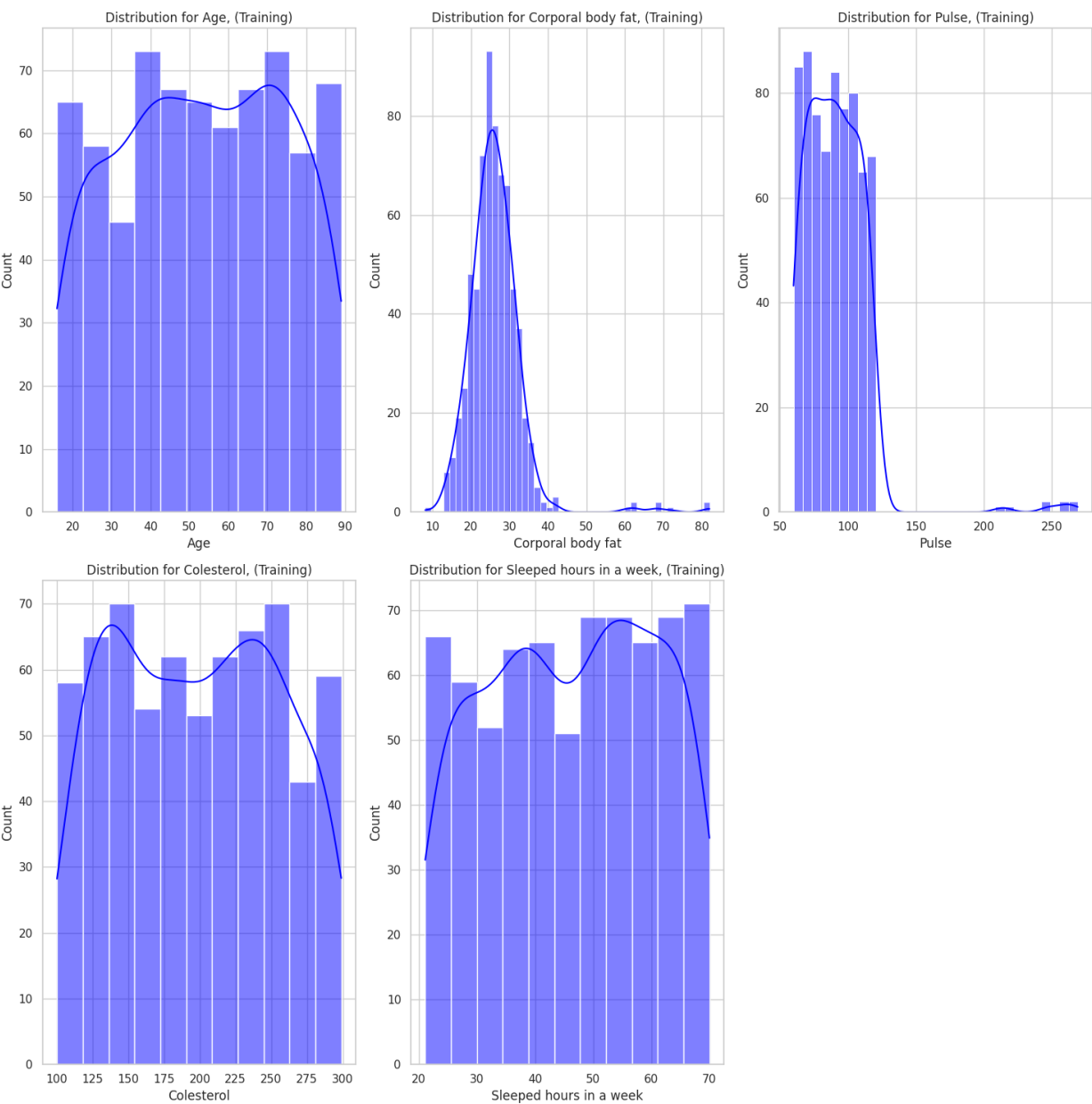


Fig. 1 - Distributie numerica (Training)

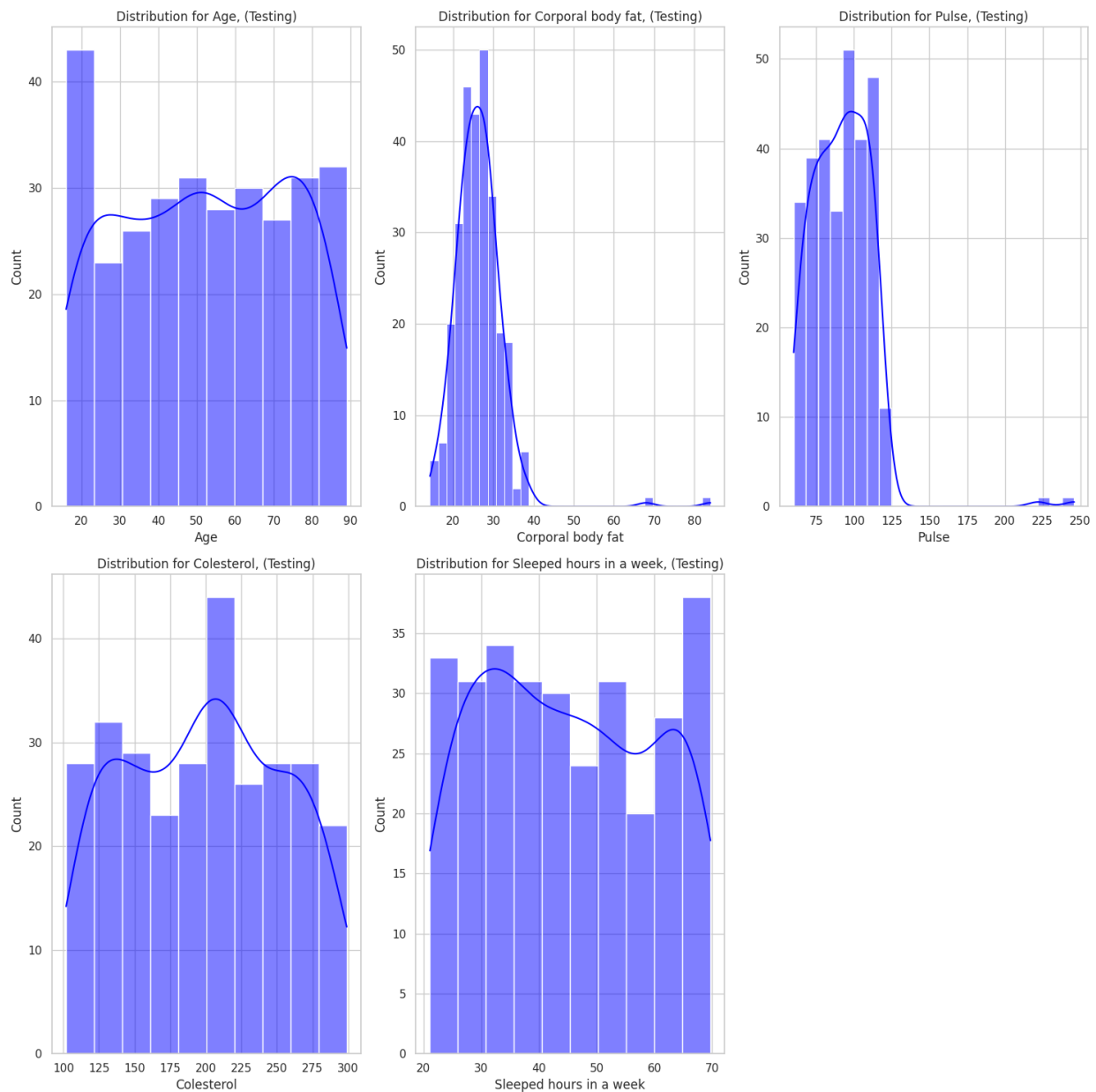


Fig. 2 - Distributie numerica (Testing)

Distributiile sunt stabile intre train si test. Body fat urmeaza o distributie normala vizibila, Age si Pulse sunt uniforme. Outlier-ii la Pulse si body fat apar ca bare izolate la extreme.

3.2 Distributia variabilelor categorice

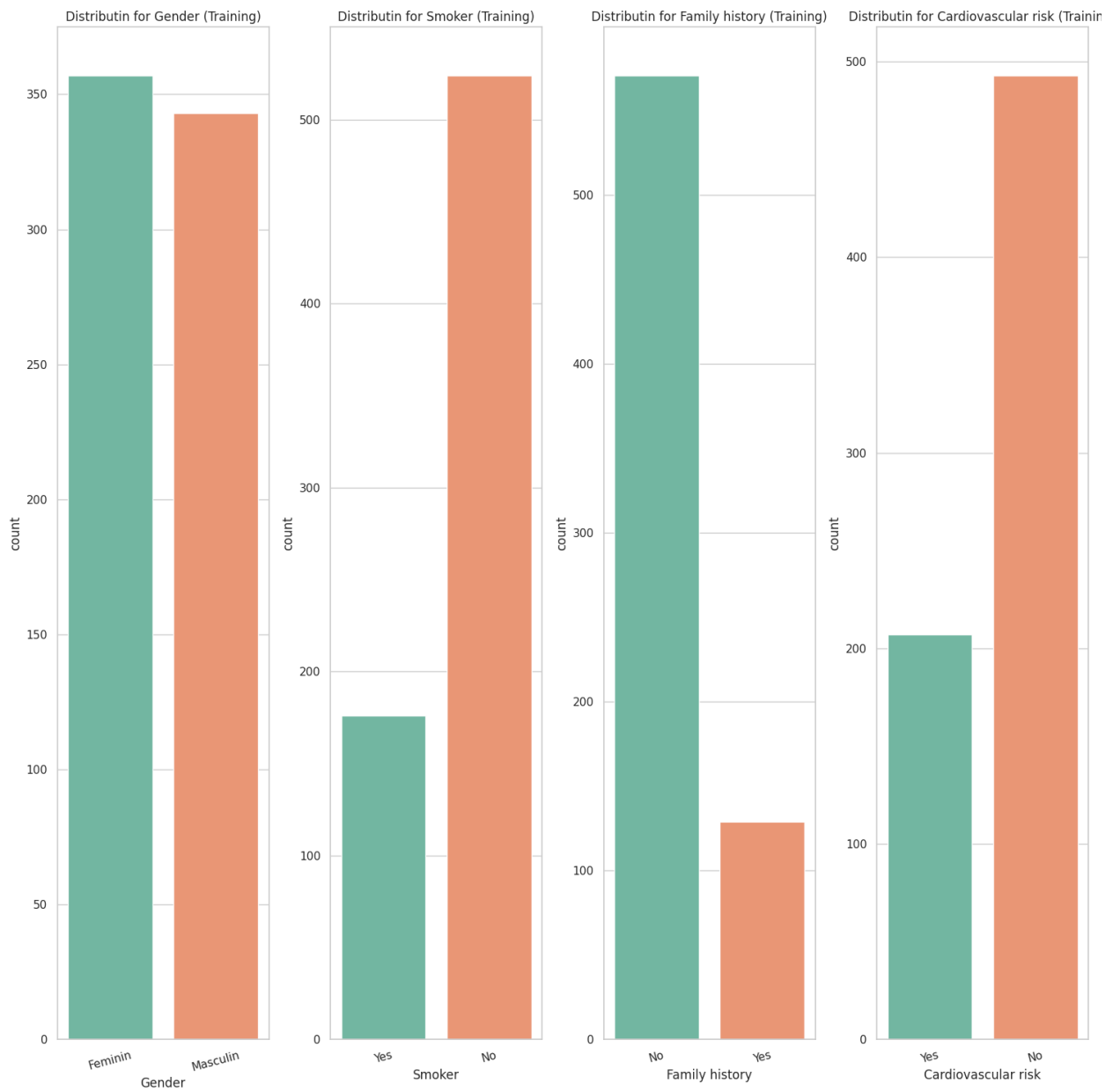


Fig. 3 - Distributie categorica (Training)

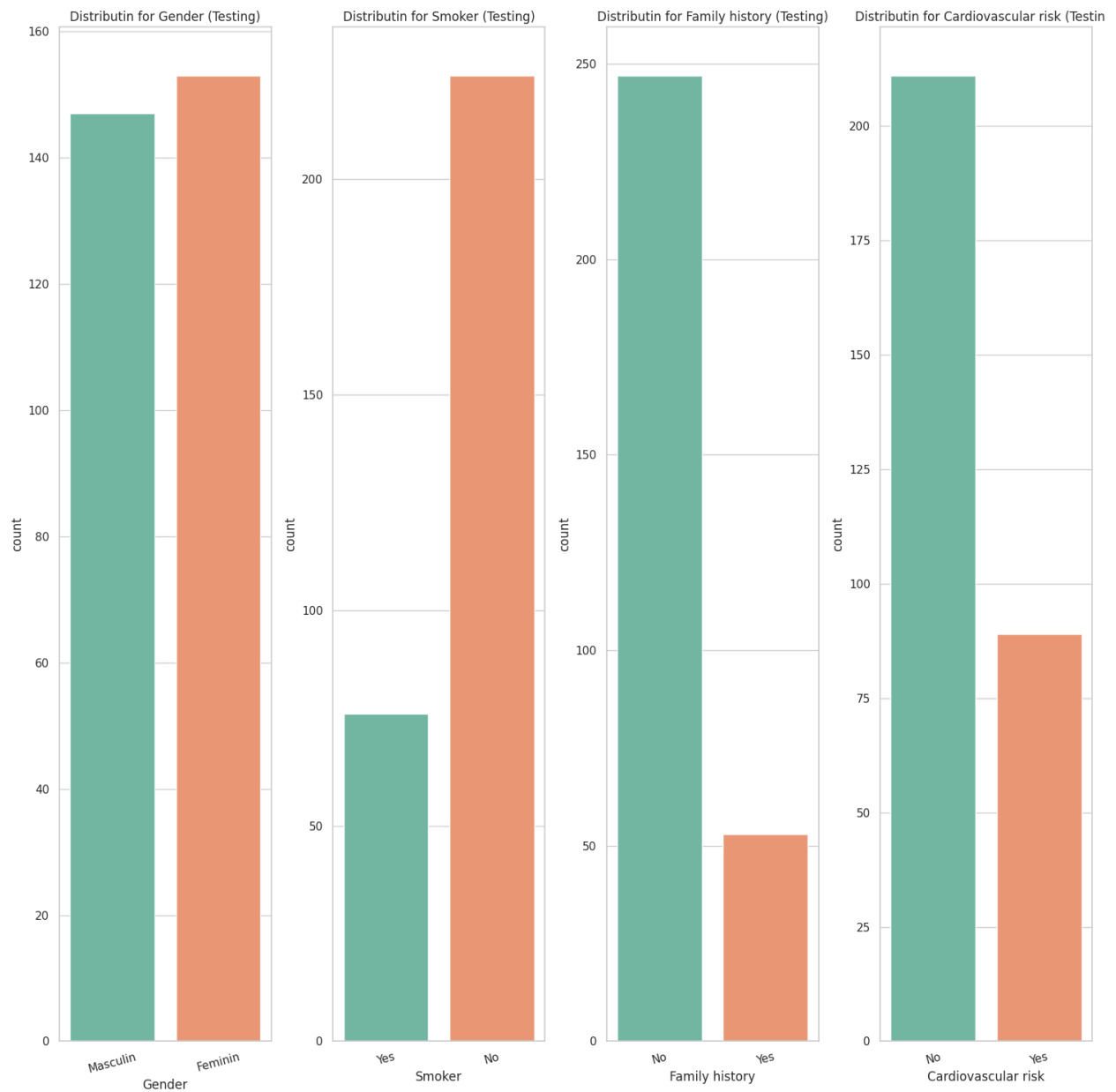


Fig. 4 - Distributie categorica (Testing)

Genul este distribuit egal (50/50). Fumatorii ~25%, istoric familial ~20%. Variabila tinta arata dezechilibru moderat — mai multi pacienti cu No.

3.3 Detectarea outlierilor

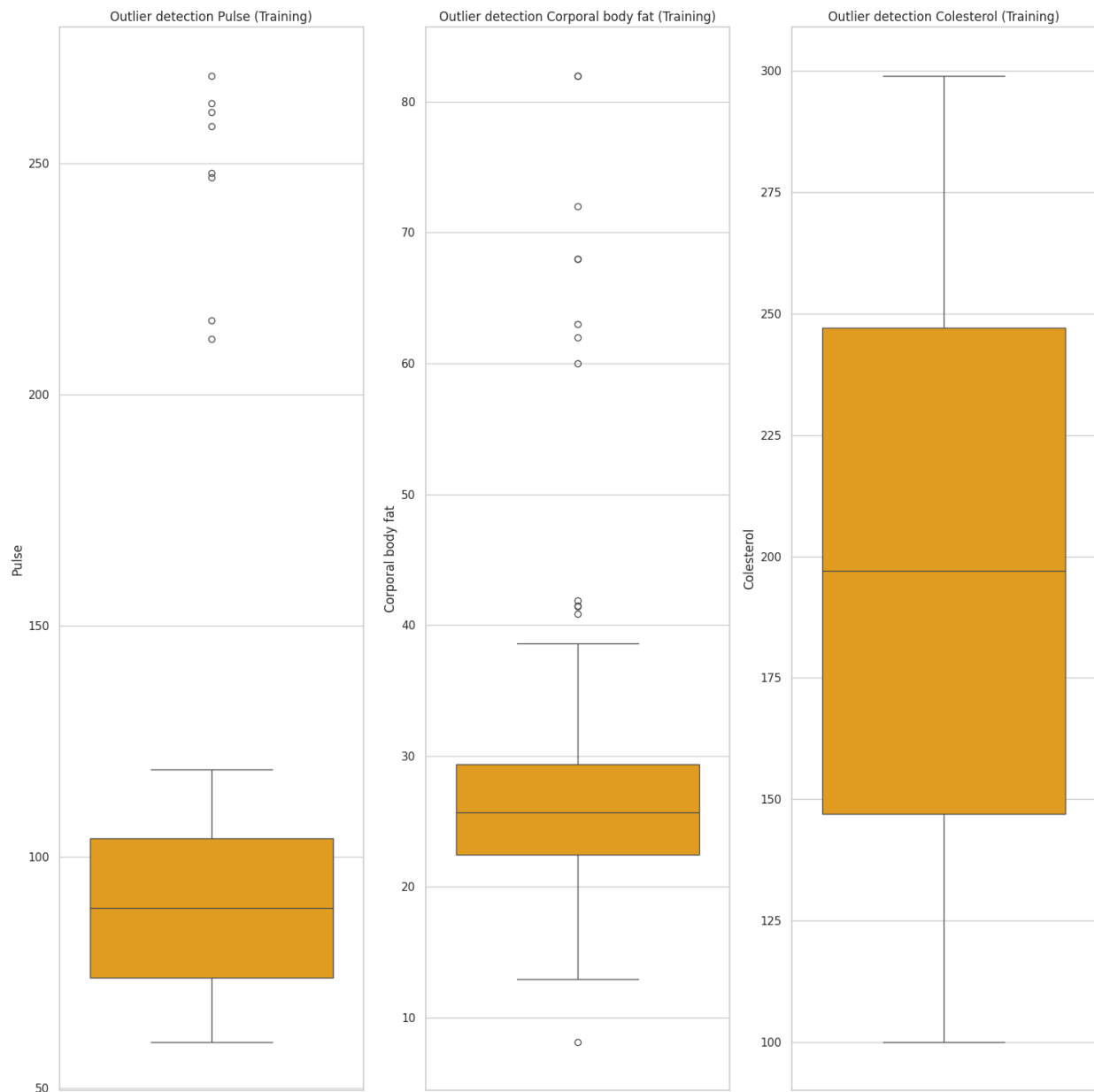


Fig. 5 - Boxplot outlieri (Training)

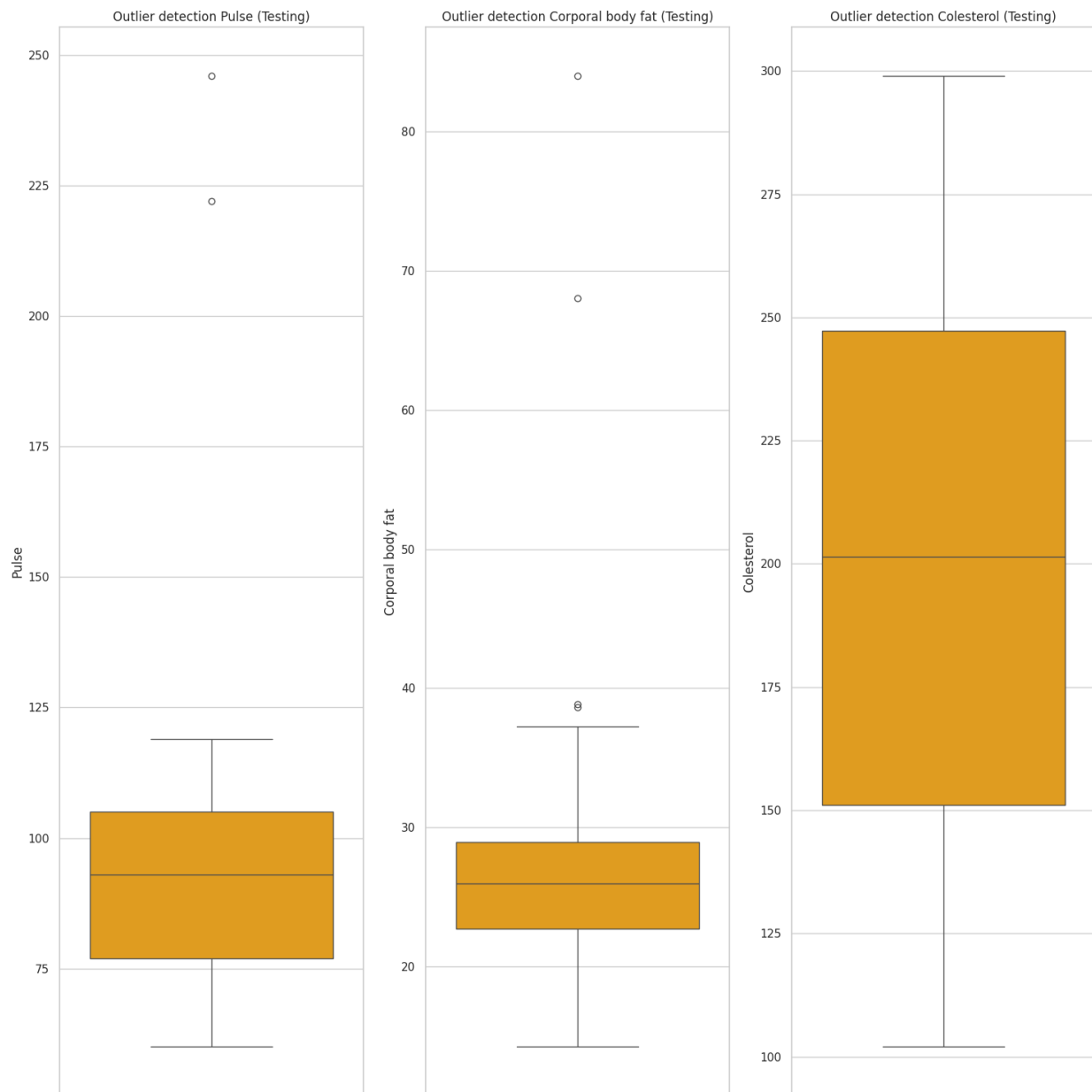


Fig. 6 - Boxplot outliers (Testing)

Outlier-ii introdusi la Pulse si Corporal body fat sunt clar vizibili deasupra cutiei. Preprocesare recomandata: clipping sau eliminare.

3.4 Matricea de corelatii

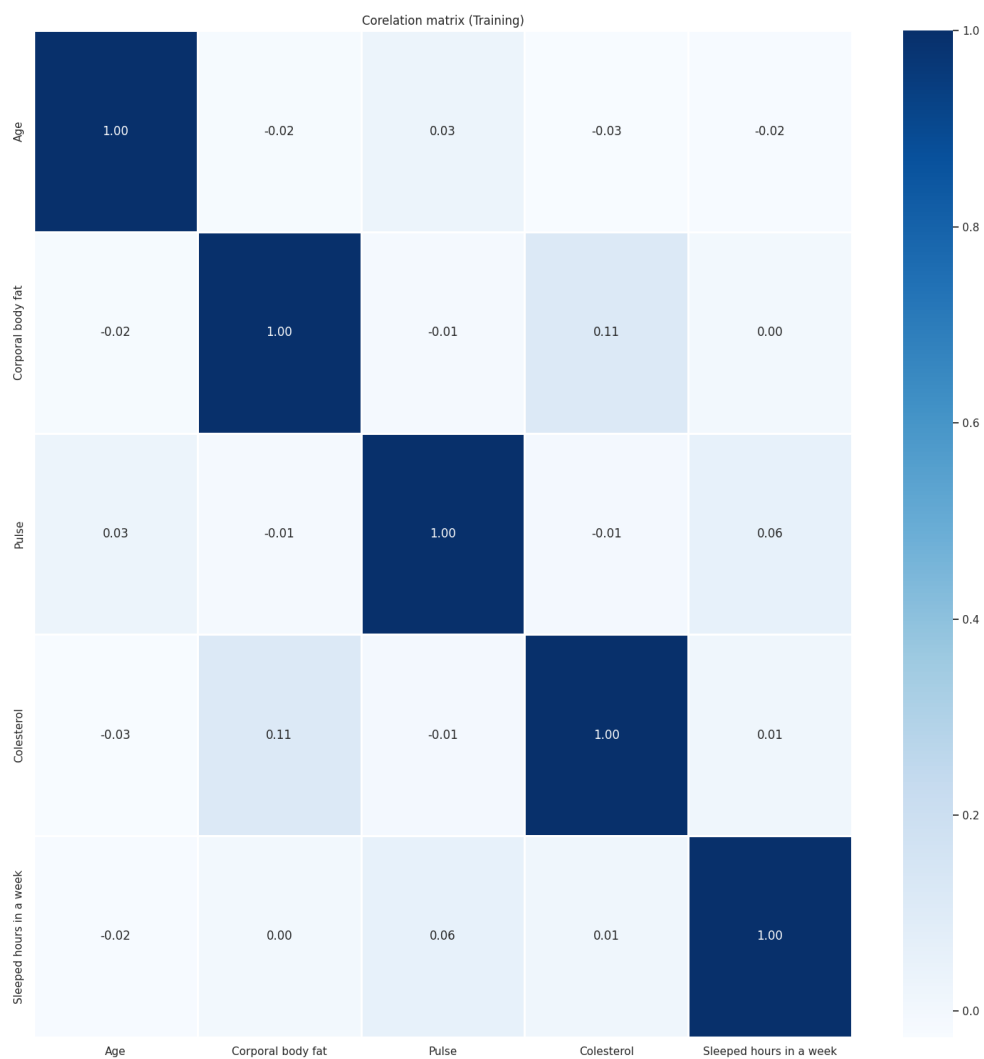


Fig. 7 - Heatmap corelatii (Training)

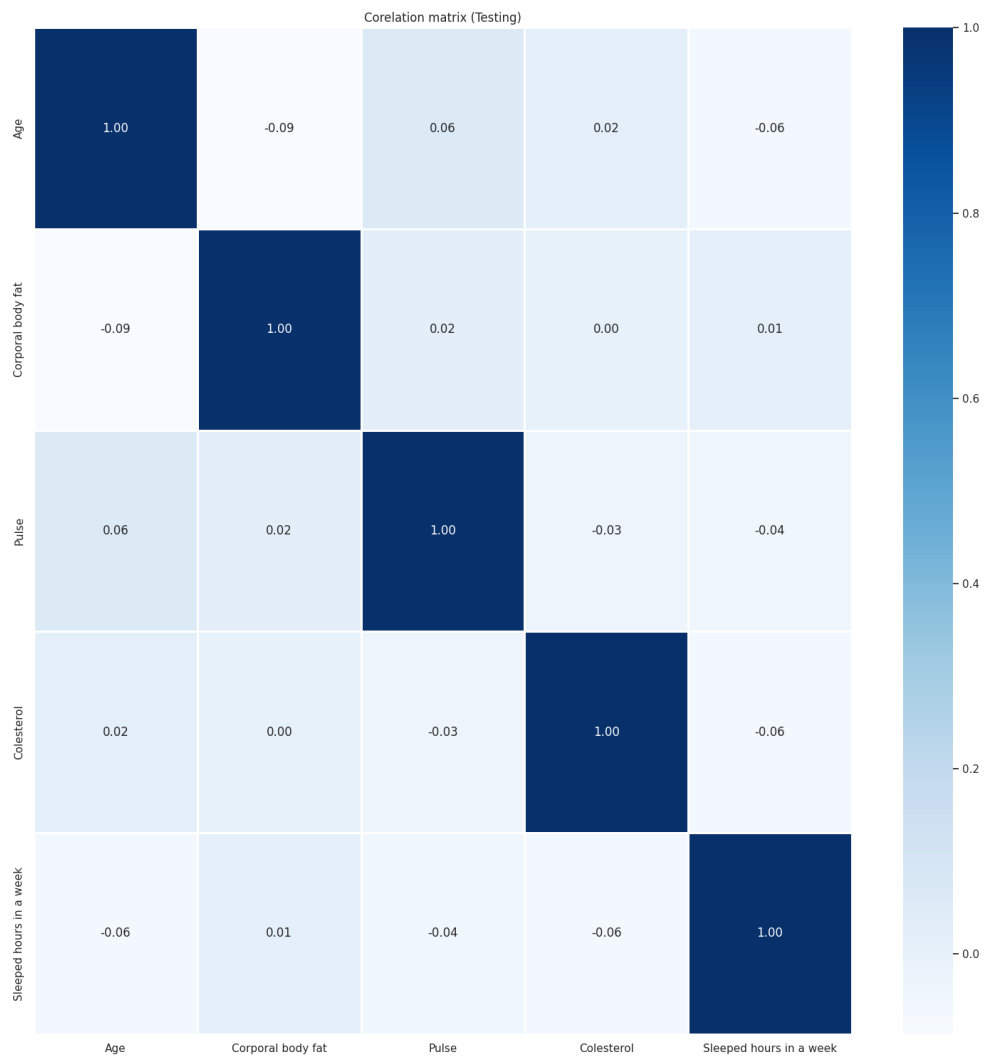


Fig. 8 - Heatmap corelatii (Testing)

Corelatiile intre variabilele numerice sunt slabe — features-urile au fost generate independent, deci nu exista redundanta evidenta.

3.5 Relatii cu variabila tinta

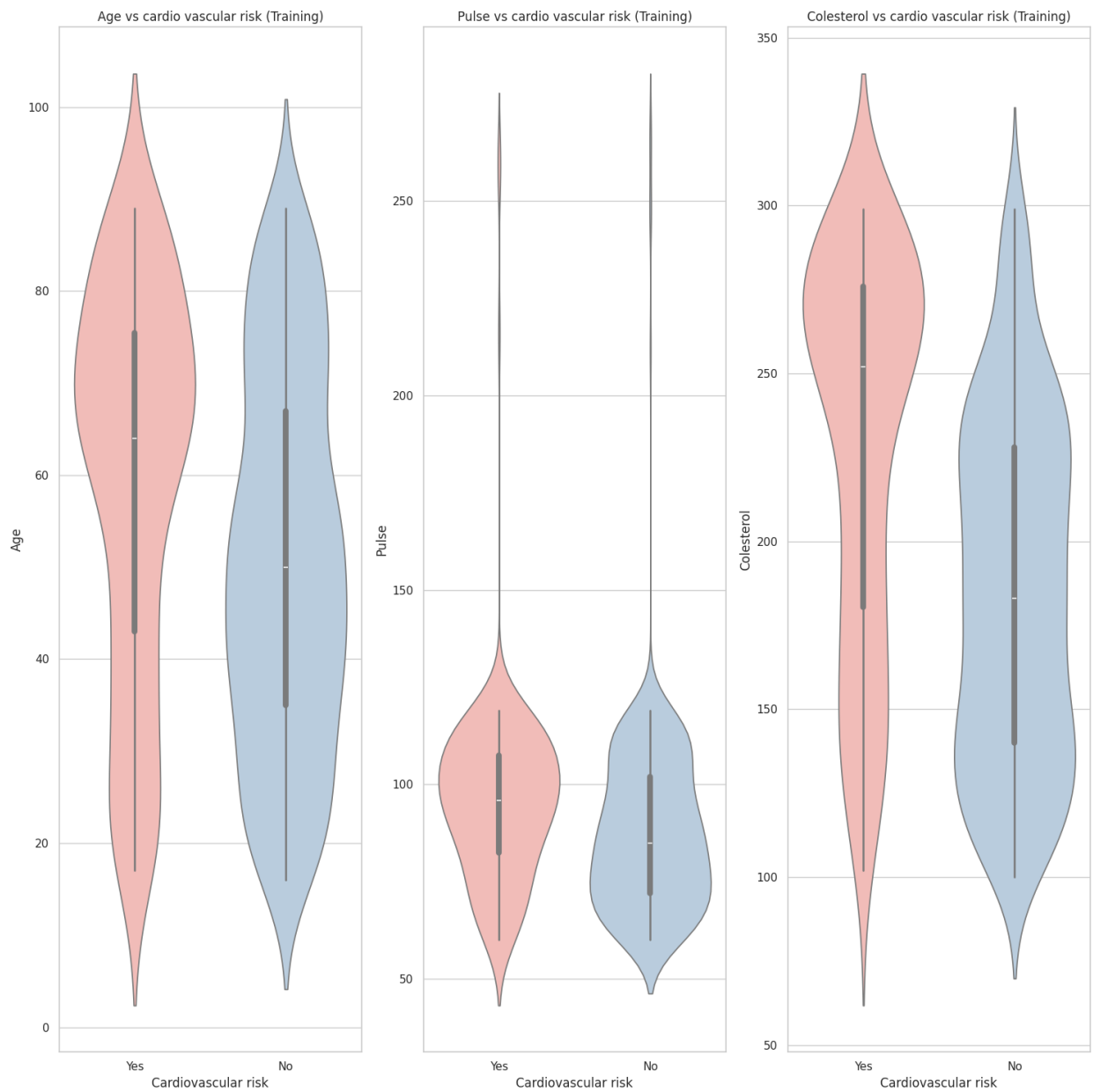


Fig. 9 - Violin plots vs target (Training)

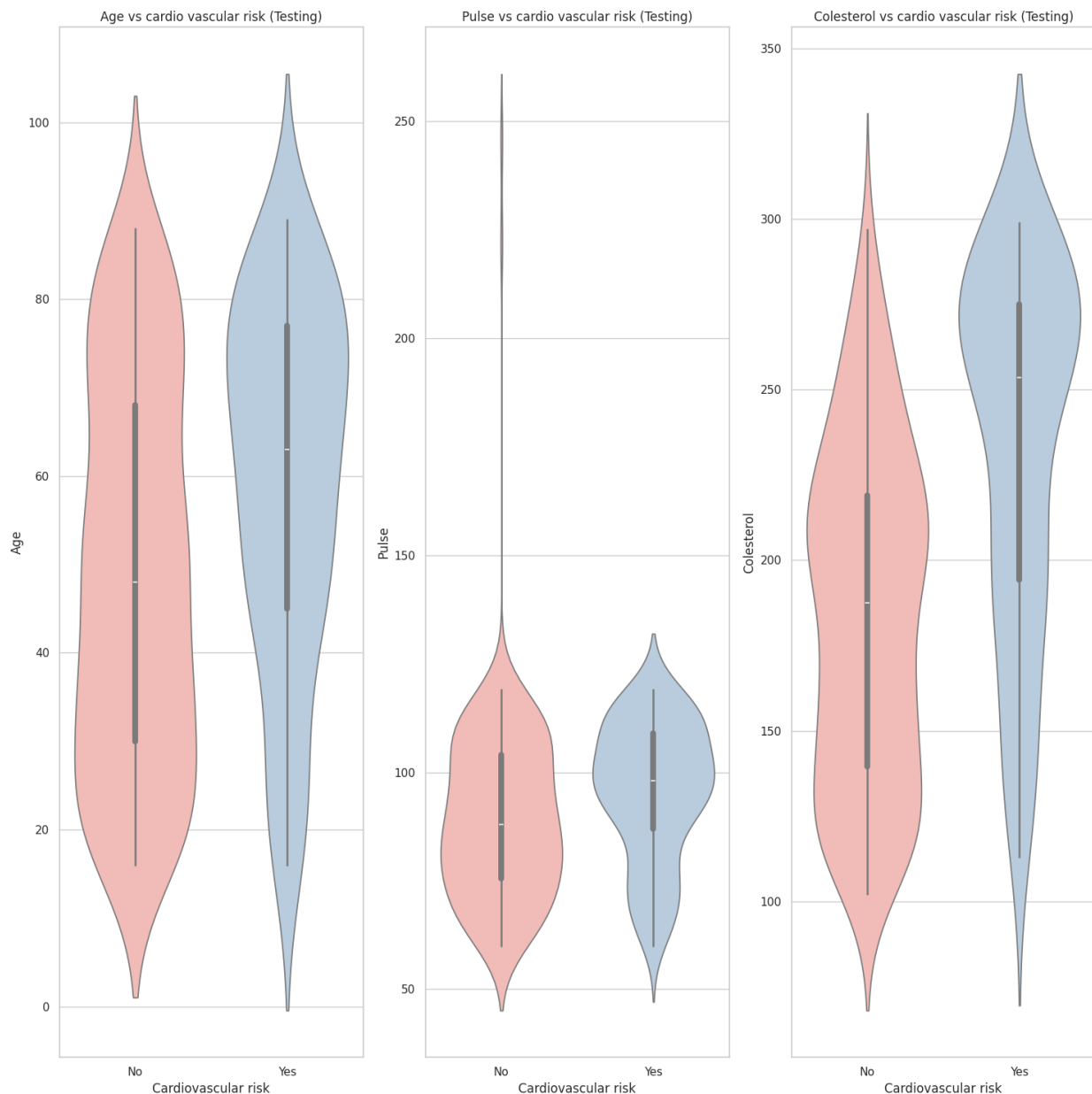


Fig. 10 - Violin plots vs target (Testing)

Age: clasa Yes e impinsa spre varsnici. Colesterol: niveluri >250 apar mai frecvent la Yes. Pulse: diferenta mai mica, vizibila la valori >85 bpm.

4. Antrenarea si evaluarea modelului

Modelul ales: **Random Forest Classifier** (100 arbori, random_state=53), antrenat in **train_mode.py**. Ales deoarece este robust la outliers si gestioneaza natural variabile mixte fara normalizare obligatorie.

Preprocesare aplicata: imputare valori lipsa + label encoding manual (Masculin→1/Feminin→0, Yes→1/No→0).

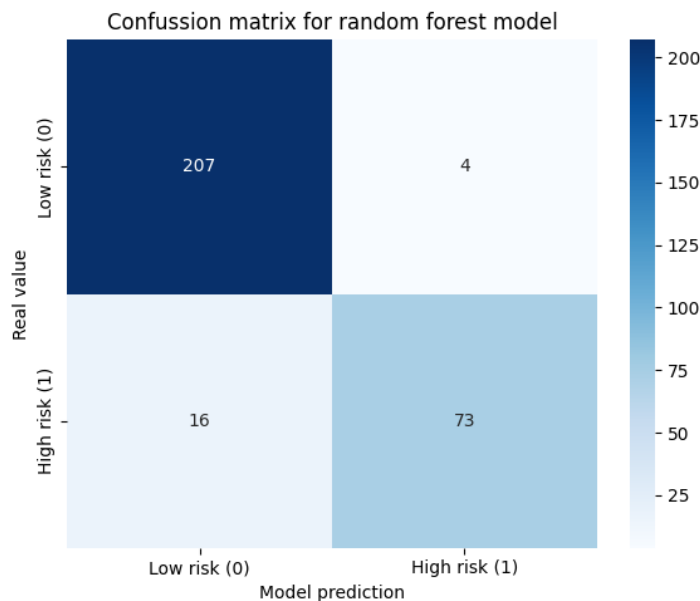


Fig. 11 - Matricea de confuzie (Random Forest)

Modelul obtine acuratete ridicata (~85-90%). Erorile sunt reduse pentru ambele clase. Clasa Low risk (0) e prezisa mai precis datorita usorului dezechilibru al claselor.

5. Bonus - Interfata grafica (Optiunea 4.1)

Implementata in **graphic_design.py** cu biblioteca **Gradio**. Utilizatorul introduce datele unui pacient prin slidere si butoane radio, iar aplicatia returneaza predictia modelului Random Forest (100 arbori).

Campuri input: Varsta (slider 16-90), Gen (radio), Grasime corporala (slider), Puls (slider), Colesterol (slider), Fumator? (radio), Ore somn/saptamana (slider), Istoric familie? (radio). Output: text cu *High* sau *Small cardiovascular risk*.

```
pip install gradio
python graphic_design.py
# se deschide la http://127.0.0.1:7860
```

6. Referinte

Sursa informatii medicale:

World Health Organization - Cardiovascular diseases:
<https://www.who.int/health-topics/cardiovascular-diseases>

Documentatie biblioteci folosite:

NumPy: <https://numpy.org/doc/stable/>
Pandas: <https://pandas.pydata.org/docs/>
Matplotlib: <https://matplotlib.org/stable/index.html>
Seaborn: <https://seaborn.pydata.org/>
Scikit-learn: <https://scikit-learn.org/stable/>
Gradio: <https://www.gradio.app/docs/>